

System Solutions

Capability Brochure

From hardware architecture to connected, production-ready systems.



Engineering across the complete system

Bring hardware, software, and connectivity together with engineering support spanning design, platform enablement, validation, and manufacturing readiness.

Hardware

Architecture, PCB design, bring-up, and validation.

Software

Firmware, BSPs, drivers, RTOS, and Linux.

Integration

Connectivity, system testing, and production support.

Capabilities that work together

Hardware Engineering

System architecture, component selection, schematic and PCB design, power design, and board bring-up.

Firmware Development

C/C++ firmware, bootloaders, peripheral integration, control algorithms, and performance optimization.

Platform Enablement

Board support packages, hardware abstraction, Linux and RTOS porting, and middleware integration.

Real-Time Systems

Task scheduling, interrupt handling, synchronization, multithreading, and real-time performance tuning.

Linux & Device Drivers

Yocto and Buildroot platforms, kernel configuration, peripheral drivers, secure boot, and OTA updates.

Connectivity & IoT

Wireless and wired interfaces, gateways, device protocols, edge-to-cloud connectivity, and edge AI enablement.

Automotive Systems

ECU development, CAN/LIN and UDS integration, diagnostics, and functional safety architecture support.

Testing & Validation

Unit, integration, and system testing; HIL, regression automation, stress testing, and CI/CD workflows.

Production Support

DFM/DFT reviews, test fixtures, programming jigs, manufacturing documentation, and production test planning.

System Integration

Hardware, software, and middleware integration with debugging and system-level validation.

Across industries

Industrial IoT / Healthcare / Automotive / Consumer electronics / Aerospace / Edge AI

Let's build your next system.

Tell us about your project: hr@vconnectech.in